JAYHAN B. DAYAG

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EDUCATION

CAGAYAN STATE UNIVERSITY CARIG CAMPUS
Bachelor of Science in Computer Engineering | 2023-Present
CATAGGAMAN NATIONAL HIGH SCHOOL
Humanities and Social Science, Senior High School | 2017-2023
CATAGGAMAN ELEMENTARY SCHOOL | 2011 – 2017

PROJECTS

- **PIC DEVELOPMENT KIT BOARD**
  All-in-one hardware platform designed to simplify the process of prototyping and testing applications using Microchip's PIC (Peripheral Interface Controller) microcontrollers.
- **POWER SUPPLY**
  End-to-end development of a DC-DC/AC-DC power supply, from schematic capture and simulation to multilayer PCB layout and hardware validation, achieving 100% efficiency.
- **GOLDEN HARVEST SUBDIVISION BOARDING HOUSE FINDER**
  A web-based management and rental platform designed to help boarders find rooms or houses for rent within Golden Harvest Subdivision while allowing landlords/homeowners to post, manage, and monitor their rental listings efficiently.
- **ECO-CHARGE**
  An Arduino-based, sustainable charging system designed to detect and collect recyclable plastic bottles, providing mobile device charging in exchange for each accepted bottle while promoting proper waste disposal and environmental. sustainability.
- **DINOSAUR MUSEUM MANAGEMENT SYSTEM**
  A relational database and web-based system designed to manage dinosaur exhibits, track specimen activities, log visitor interactions, and streamline museum operations.

PROJECT EXPERIENCES

- Designed and developed a custom PCB power supply, utilizing MULTISIM for circuit design and simulation, while managing the complete hardware lifecycle—from layout and fabrication to rigorous testing and performance validation
- Gained comprehensive hands-on experience in PIC microcontroller architecture by developing and programming embedded projects using C and Assembly within the MPLAB X IDE, utilizing PICKit for hardware-level debugging and system validation on development kit boards.
- Performed test case creation and execution to identify bugs, system errors, and UI/UX inconsistencies. Reported issues to the development team and verified fixes through regression testing.
- Gained hands-on experience in network installation, configuration, and troubleshooting. Assisted with NAP installation and testing, modem and router configuration, fiber optic splicing, and optical power meter testing. Developed practical skills in telecommunications infrastructure, network connectivity, equipment handling, and technical support.
- Created and executed test cases to identify bugs, system errors, and UI/UX inconsistencies. Documented and reported issues to the development team, verified implemented fixes, and performed regression testing to ensure system functionality, reliability, and usability.

SKILLS

- **Hardware**

  I have skills in hardware and embedded engineering, including working with PIC microcontrollers, MPLAB X IDE, and PICKit debugging tools. I am familiar with circuit simulation using MULTISIM, PCB design and layout, circuit troubleshooting, and hardware diagnostics. These skills help me understand electronic systems, identify technical issues, and develop reliable hardware solutions.

- **Software**

  I have knowledge of software and web development, including programming in Embedded C, Assembly language, and Python, as well as JavaScript and TypeScript. I also work with Vue.js for frontend development, Django REST Framework for backend development, and HTML5 and CSS3 for building responsive and functional websites. I am interested in developing applications that integrate software, databases, and hardware technologies to solve real-world problems.

- **Core Engineering Competencies**

  I possess analytical and problem-solving skills that help me identify issues and develop effective technical solutions. I can collaborate with team members, understand system architecture involving frontend, backend, and database integration, and perform quality assurance tasks such as test case execution, bug reporting, and regression testing. I am committed to improving system functionality, reliability, and overall user experience through systematic testing and continuous learning.

CIVIL SERVICE PROFESSIONAL ELIGIBLE (MARCH 2025)